UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING (UCSC)

SCS2301 / IS2201: Software Engineering Project • Academic Year 2025/2026

Interim Action Plan Document

Project: University Sports Management System (USMS) | Group: IS03 | Proposal Outcome: ACCEPTED

1. Executive Summary

During the proposal presentation and defense for Group IS03, the evaluation panel accepted our project proposal. The overall outcome was recorded as ACCEPT. The panel provided constructive feedback on five specific areas: handling student verification without live database access, treating security as an explicit functional requirement, ensuring multi-sport support across all university sports, removing automated medical locks, and revising our rejected design diagrams. This document clearly outlines the feedback received, explains what it means, and details the actions we have taken or plan to take.

2. Summary of Feedback Received

Evaluation Area	Panel Feedback / Comment	Current Status
1. Student DB Verification	*You can't cross-check with the student database practically. So, it is okay to assume that the verification against real university students can be done when such information is available.*	**Addressed (Internal Verification Queue)**
2. Security Classification	*Security is a functional requirement.*	**Addressed (6 Security Functional Requirements)**
3. Sports Coverage	*You mentioned that you will be covering all the sports.*	**Addressed (Dynamic Multi-Sport Catalog)**
4. Medical Lock Release	*You don't have to automate medical lock release as it may result in unnecessary complications.*	**Addressed (Manual Clinical Clearance)**
5. Design Diagrams (Reject)	*Standardize component diagrams and ensure UI/component consistency across team branches on GitHub.*	**Rectified (UML Component Diagram & Shared UI)**

3. Detailed Feedback & Actions Taken

3.1 Student Database Verification (Assumption Modeling)

Feedback Received: "You can't cross-check with the student database practically. So, it is okay to assume that the verification against real university students can be done when such information is available."

What this means: As 2nd-year undergraduate students, we do not have access to live University of Colombo student administration APIs due to privacy regulations. The panel acknowledged this practical reality and permitted us to treat external university database verification as an assumption.

Action Taken / Plan:

- Built an Internal Verification Queue in the Admin portal (views/admin_user_management.php).
- When a student registers as an athlete, they submit their university Index Number and Faculty. Their profile is placed into a 'Pending' verification queue.
- An Admin or TIC officer reviews the registration details and clicks 'Approve' or 'Reject'.

- Prepared an extensible service adapter interface so that when official institutional API access is granted, the live connection can be plugged in without changing the core codebase.

3.2 Security as a Functional Requirement

Feedback Received: “Security is a functional requirement.”

What this means: The evaluators highlighted that security must not be treated as a simple non-functional bullet point. It must be built as working functional features that can be tested and demonstrated.

Action Taken / Plan:

- BCRYPT Password Hashing: All user passwords are encrypted using PHP password_hash(..., PASSWORD_BCRYPT) and verified using password_verify() in models/User.php and controllers/AuthController.php.
- Role-Based Access Control (RBAC): A central middleware guard (config/auth_guard.php) strictly enforces permissions for 8 roles (Admin, Coach, TIC, Medical Officer, President, Athlete, etc.). Unauthorized page access is blocked automatically.
- SQL Injection Prevention: All database queries across all models use PDO Parameterized Prepared Statements (PDO::ATTR_EMULATE_PREPARES => false).
- XSS Protection: All dynamic output is escaped with htmlspecialchars(..., ENT_QUOTES, 'UTF-8').
- Audit Logging: Created views/admin_audit_logs.php to record sensitive administrative actions (user edits, deletions, role updates).

3.3 Sports Coverage (Supporting All University Sports)

Feedback Received: “You mentioned that you will be covering all the sports.”

What this means: In the proposal, our group committed to supporting all university sports, not just one sport. The panel reminded us to ensure our architecture dynamically accommodates multiple sports.

Action Taken / Plan:

- Designed a centralized sports database table that supports all 24+ recognized University of Colombo sports (Cricket, Badminton, Football, Basketball, Swimming, Track & Field, Rugby, Tennis, etc.).
- All operational tables—training_sessions, fixtures, match_squads, equipment, and injuries—are relationally mapped to sport_id.
- Built an administrative Sports Catalog view (views/admin_sports_catalog.php) allowing administrators to add, view, or configure any sport dynamically without modifying any code.

3.4 Medical Lock Release (Manual Clinical Sign-Off)

Feedback Received: “You don't have to automate medical lock release as it may result in unnecessary complications.”

What this means: If an athlete gets injured and the system automatically unlocks them after 14 days, an unfit player could compete and suffer severe injuries. The panel emphasized that medical clearance must strictly be a doctor's decision.

Action Taken / Plan:

- Completely removed any automated or time-based medical lock release.
- Made medical clearance 100% manual: only an authorized Medical Officer can certify that an athlete is fit to compete (views/med_clearance.php and models/Medical.php).
- The Medical Officer must clinically examine the athlete, enter medical assessment notes, and manually submit the clearance certification.

3.5 Design Diagrams (Rectifying Rejection & Component Consistency)

Feedback Received: Design Diagrams were marked as 'Reject' in the evaluation spreadsheet.

What this means: The early design diagrams lacked standard UML formatting, had architectural discrepancies, and showed inconsistent UI components across different team members' branches.

Action Taken / Plan:

- Standardized our UML Component Diagram into a clean 3-tier MVC architecture (Dynamic Views -> Access Guard Middleware -> Controllers -> Models -> MySQL Database).

- Unified the UI Design System across all roles on GitHub (Hemsara04/USMS): established consistent colors (Deep Navy #07162D, Athletic Teal #087F80), shared navigation sidebars, standardized metric cards, and clean data tables.
- Currently updating the Activity and Sequence diagrams to match the exact implemented code before the final presentation.

4. Action Plan Timeline & Milestones

To ensure all components are completed on time for the final defense, we have organized our work into clear milestone sprints:

Sprint / Milestone	Key Tasks & Deliverables	Status
Sprint 1 (Baseline)	Unified login portal, BCRYPT password hashing, RBAC middleware, MySQL schema setup.	**Completed**
Sprint 2 (Interim)	Admin portal User Management CRUD, verification queue, dynamic dashboard statistics.	**Completed**
Sprint 3 (Current)	TIC Equipment & Stock Inventory CRUD, checkouts, Medical Officer manual clinical clearance.	**Completed**
Sprint 4 (Diagrams)	Overhaul UML Component, Activity, and Sequence diagrams to match the live MVC code.	**In Progress**
Sprint 5 (Quality Assurance)	UI consistency audit across student/coach portals, responsive testing, edge-case validation.	**Planned**
Final Phase	Final documentation, deployment preparation, video demonstration, and system defense.	**Scheduled**

5. Team Members & Contributions

All features are tracked and integrated on our shared GitHub repository: Hemsara04/USMS (main branch).

• **Hasitha Hemsara** (Index: 22020294 / IS) — Role: Backend & Security Lead (Authentication, RBAC, Admin & TIC CRUD)

• **Sandaruwan Bandara** (Index: 22020308 / IS) — Role: Coach & Match Coordinator (Training Sessions, Squad Formulations)

• **Abdullah Farook** (Index: 22020112 / IS) — Role: UI/UX & Frontend Lead (Interface Layouts, Shared Design System)